r/askscience -8 л.
gremmer_is_key

What are the advantages/disadvantages of a ternary computer vs a binary computer? Why are almost all modern computer binary computers?

23 · 25

Holy_City • 8 л. назад

u/cantgetno197 provided a great physical answer, but I'll talk a little more about the history and theory.

First off - not all computing systems are strictly binary. The Intel 8087 is a good example, which used quarternary (base 4) digits in the ROM. Even today not all digital systems use binary, telecoms are a good example.

In the theory of computing, it doesn't matter what your base is. All that matters is that you have a discrete set of unique values, called symbols. Where the choice of base comes in is when you decided how to physically represent those symbols and the hardware that is manipulated by them to manipulate other symbols.

Binary is a good choice of base because it's easy to implement in hardware, both physically with electronics and in the design of the circuits they control. We use a mathematical framework called Boolean Algebra to design the low level switching circuits manipulated by bits. All digital circuits are designed using Boolean algebra, which is based on statements being TRUE or FALSE (in binary, 1 or 0). Furthermore, logic gates (the hardware that implements boolean logic) are remarkably simple circuits, and can be built with as few as a single or pair of transistors.

Now on top of that - we have six decades of engineering know-how (give or take) building semiconductor circuits to implement Boolean logic. We can make them extremely cheap, small, and power efficient. We also have a debt - where any new system *must* be able to integrate with older systems, or else it won't be economically viable. This is why PSK and quarternary systems work while ternary doesn't. They provide the benefits but are easy to convert to binary.

15

paolog • 8 л. назад

> All digital circuits are designed using Boolean algebra, which is based on statements being TRUE or FALSE (in binary, 1 or 0). Furthermore, logic gates (the hardware that implements boolean logic) are remarkably simple circuits, and can be built with as few as a single or pair of transistors.

 e logic circuits gives us a circuit that can add two bits, and from there it's just a few steps to implementing subtraction, multiplication and division. These circuits would be a lot more complicated in ternary.

6

shy_cthulhu • 8 л. назад

Ternary logic makes my head hurt. Random example: how many different logic gates are possible with two inputs and one output? For binary, it's 16. For ternary, 19,683. No thanks.

cantgetno197 • 8 л. назад

A common engineering rule of thumb for computing semiconductor devices is that the ratio between the amount of current a device draws when "ON" versus what it draws when "OFF" needs to be greater than 10^6 or one MILLION in order for circuit designs to be able to "work" with it. In an ideal device, the off current should be zero but in reality there are always parasatic, unwanted currents.

So, my point is, you have to remember that computers AREN'T digital, they are in fact enormous ANALOG circuits made of MOSFET transistors and diodes with literally billions of elements. In principle, every transistor should be identical, in reality manufacturing inhomogeneities make the characteristics of each of the billion devices have slightly different random characteristics. In principle, there should only be "ON" currents, in reality the difference between the two is a matter of magnitude and your 0 or 1 digital signal is actually a complex, noisy analog signal.

So, engineers need a literal factor of a million difference to differentiate a binary state in light of these imperfections acting to "muddy" your signal. If you try to make a TRINARY device, which instead of current "ON" and current "OFF" has current "OFF", current "MID" and current "HIGH", you're looking at a dramatic need to reduce noise and random manufacturing variations to make those clearly distinguishable. It also goes against the natural way our semiconductor devices work, which is to "turn on" exponentially above a certain "threshold voltage". Such behaviour lends itself naturally to a 2 current conceptualization but not to a three.

41

tuctrohs • 8 л. назад

The 10^6 should be clarified. That's not really a necessity. In the ~70s, the fastest computers were made with "emitter coupled logic" which was much more sloppy in that respect, for example. The usefulness of 10^6 in modern computer circuits is minimizing power consumption of the inactive components, not so much ensuring noise margins.

15

MiffedMouse • 8 л. назад

This post is wrong.

10^6 number you quote is very misleading. While it is true that a large difference between an "on" current and an "off" current is good, bit states are not defined by currents, they are defined by voltages.

I actually know some people who have worked on ternary computer designs. For ternary designs, they divide the voltage range into sections. The rough rule of thumb they quoted is a rule of fifths. In other ... fifths, where three states are valid and the other two

...simple extension of the rule of thirds you will occasionally see for binary computers (that is, you should avoid using the middle third of the voltage range because that is where bit-flip errors become most likely).

Ternary computers do NOT try to define multiple "on" current states. Instead, feedback loops are used to drive the output to the desired voltages (just like in any other circuit). But instead of two voltage levels, now there are three output levels.

PS: even in binary circuits, bit states are not defined in terms of currents. To go from an ON state to an OFF state the current charge stored in a flip-flop (or other storage device) must be drained. This requires current. The only difference is the OFF current drives the stored voltage LOW while the ON current drives the stored voltage HIGH. (Actually, in many computers this is flipped because high versus low is arbitrary and in some devices it is easier to set it to default high, but default off is also a desirable feature, so high=off is chosen. Consult device APIs!)

PPS: While I am by no means an expert, the actual reason ternary computers are rare is that they are much more difficult to design for very little benefit. Logic gates are more difficult to design, especially because a lot of research into efficient binary logic gates has been done already while comparatively little research into ternary gates has been performed. Furthermore, even well designed ternary logic gates often require more transistors because they must be able to control for more voltage levels. The primary benefits of ternary logic are in less busses or storage space required per bit. However, this can be more easily designed and handled as a separate subsystem with converters to and from ternary, rather than changing the basis of the entire chip just to accomodate the bus.

8

jakejakejake86 • 8 л. назад

so why cant we use +5,0-5 (i know vcc is way lower now but still)

2

[удалено] • 8 л. назад

Еще 6 ответов

NotWithoutIncident • 8 л. назад

Brian Hayes has a famous article about this.

> Why did base 3 fail to catch on? One easy guess is that reliable three-state devices just didn't exist or were too hard to develop. And once binary technology became established, the tremendous investment in methods for fabricating binary chips would have overwhelmed any small theoretical advantage of other bases.

Basically the argument is that early computer pioneers didn't recognize the information theoretic advantages of ternary (the most efficient base system, being closest to e as u/Kered13 explained) and by the time people did so much money and time had been invested in binary computers that they more than made up for any advantage ternary systems might have.

5

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r/musictheory • -8 л.

Binary vs Ternary. I haven't found a single explanation that makes me understand it. Please help.

6 13

r/computers • -2 г.

r/answers • -8 л.

Why are we using binary computers instead of ternary computers?

4 10

r/worldbuilding • -6 л.

How would ternary computers differ from binary computers?

7 9

r/todayilearned • -10 л.

TIL there have been computers programmed in "ternary" instead of the conventional binary. One such computer, built by the Soviets in 1958, had distinct advantages over binary...

5,3 тыс. 301

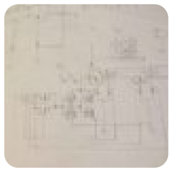
r/ProgrammerHumor • -4 г.

Ternary operators are the worst..unless you indent them right!

3,2 тыс. 328

r/computerscience • -1 г.

I designed my own ternary computer



175 45

r/csharp • -3 г.

Why I don't like ternary operator? Make a quick change to this :D

0 44

r/compsci • -5 л.

What's the difference between a ternary computer and a quantum computer?

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r/learnprogramming • -4 г.

Is using the ternary operator bad practice?

1 11

r/math • -10 л.

r/QuantumComputing • -6 л.

Is a quantum computer a ternary computer?

4 27

r/hardware • -5 л.

Trinary computing and beyond - A way to break through performance limitations of today's Binary systems?

17 15

r/computers • -17 дн.

Ternary system

1 8

r/askscience • -15 л.

I'm no computer expert, so forgive me if this is a ridiculous question: What would happen if computers switched to trinary instead of binary?

196 116

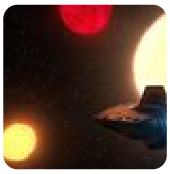
r/PowerShell • -1 г.

if statement vs. ternary operator

16 29

r/EliteDangerous • -10 л.

Is "trinary" a word?



38 28

r/cpp • -11 л.

C++ ternary operator considered (very) harmful

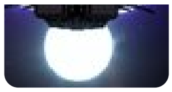
0 11

Пропустить и перейти к основному содержимому

…arent access point?

241 63

Trinary system? Is that a thing?



171 50

r/learnrust • -4 г.

Ternary Operator or Equivalent?

20 16

r/programming • -11 л.

"Perhaps the prettiest number system of all," writes Donald E. Knuth in The Art of Computer Programming, "is the balanced ternary notation."

750 179

r/programming • -17 л.

Russian Ternary Computer

174 60

r/programming • -17 л.

Three Cheers for Base 3: A Quick Introduction to the Advantages of Ternary Notation.

96 58

r/compsci • -2 г.

What numeral system would work well with a Trenary computer

13 30

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